

Locations in a city are vertically differentiated by quality θ . θ can be determined by multiple characteristics, but there is agreement on the ranking. At a location with quality θ , price per square foot of a home in an Nplex (a collection of N homes), each of which has square footage $s(N)$ is $p(N, \theta)s(N)$. We assume that builders are always at a corner solution of building up to $s(N)$ square feet per unit. The function $p(N, \theta)$ includes the relationship between size and number of units implicitly in N . The cost per square foot of an Nplex is $c(N)$, with $c' \geq 0$ because of relatively fixed plumbing and utility costs and fire separation.

The profit from building an Nplex at location θ is thus:

$$\pi(N, \theta) = [p(N, \theta) - c(N)] s(N) \quad (1)$$

Treating the number of units as continuous:

$$\frac{\partial \pi(N, \theta)}{\partial N} = [p(N, \theta) - c(N)] s'(N) + \left[\frac{\partial p(N, \theta)}{\partial N} - c'(N) \right] s(N). \quad (2)$$

Differentiating with respect to θ :

$$\frac{\partial^2 \pi(N, \theta)}{\partial N \partial \theta} = \underbrace{\frac{\partial^2 p(N, \theta)}{\partial N \partial \theta} s(N)}_{\text{division effect}} + \underbrace{\frac{\partial p(N, \theta)}{\partial \theta} s'(N)}_{\text{addition effect}} \quad (3)$$

The sign of the first, the “division effect” term depends on income sorting at home size N . If higher income households live in better locations at N units per lot, then we would expect $\frac{\partial^2 \pi}{\partial N \partial \theta} < 0$, as sharing space and smaller spaces are inferior goods. The second term, the “addition effect” is positive if $s'(N) > 0$, that is, if choosing a larger number of units allows more square feet to be built.